Masked observations

1 Masked observations

Theorem 1.1. *For every natural number t,* $\mathrm{whitecount}(t, 0) = 0$.

Proof. The goal follows from $\mathrm{whitecountzero}(t)$. □

Theorem 1.2. *For every natural number t and natural number n, if* $\mathrm{center}(t) = \mathrm{true}$*, then* $\mathrm{whitecount}(t, n + 1) = \mathrm{whitecount}(t + 1, n)$.

Proof. The goal follows from $\mathrm{whitecountblackstep}(t, n)$. □

Theorem 1.3. *For every natural number t and natural number n, if* $\mathrm{center}(t) = \mathrm{false}$*, then* $\mathrm{whitecount}(t, n + 1) = \mathrm{whitecount}(t + 1, n) + 1$.

Proof. The goal follows from $\mathrm{whitecountwhitestep}(t, n)$. □

Theorem 1.4. *For every natural number $t1$ and natural number $t2$ and natural number n, if* $\mathrm{centerwindowagree}(t1, t2, n)$*, then* $\mathrm{leftprefixagree}(t1, t2, n)$ *if and only if* $\mathrm{whitetraceagree}(t1, t2, n)$.

Proof. The goal follows from $\mathrm{prefixagreementiffwhitetrace}(t1, t2, n)$. □

Theorem 1.5. *For every natural number T and natural number P and natural number t and natural number n, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then there exists a natural number a such that there exists a natural number b such that $a < b$ and $b \leq$* $\mathrm{twopow}(\mathrm{whitecount}(t, n)) \wedge \mathrm{leftprefixagree}(t + a \cdot P, t + b \cdot P, n)$.

Proof. The goal follows from $\mathrm{compressedprefixrepeat}(T, P, t, n)$. □

Theorem 1.6. *For every natural number T and natural number P and natural number t and natural number n and natural number M, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then if* $\mathrm{prefixseparated}(t, P, n, M)$*, then* $M \leq \mathrm{twopow}(\mathrm{whitecount}(t, n))$.

Proof. The goal follows from $\mathrm{separatedprefixbound}(T, P, t, n, M)$. □

Theorem 1.7. *For every natural number T and natural number P and natural number t and natural number n, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then* $n < t + \mathrm{twopow}(\mathrm{whitecount}(t, n)) \cdot P$.

Proof. The goal follows from $\mathrm{edgeinformationbound}(T, P, t, n)$. □

Theorem 1.8. *For every natural number n and natural number t and natural number w and natural number P, if $n < (w+1) \cdot P$, then $n < t + \mathrm{twopow}(w) \cdot P$.*

Proof. The goal follows from $\mathrm{informationboundaudit}(n, t, w, P)$. □

2 Central collision recurrence

Theorem 2.1. *For every natural number Q and natural number t and natural number r, if $\mathrm{leftadjacentdefect}(Q, t) = \mathrm{true}$, then if $\mathrm{center}(t + r) = \mathrm{true}$, then if $\mathrm{center}(t + r + Q) = \mathrm{true}$, then $\mathrm{nearcollision}(Q, t, r + 1, 2 \cdot (r + 1))$.*

Proof. The goal follows from $\mathrm{unitdefectforcesnearcollision}(Q, t, r)$. □

Theorem 2.2. *For every natural number T and natural number P and natural number m and natural number t, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then if $T \leq t$, then if $\mathrm{leftadjacentdefect}(m \cdot P, t) = \mathrm{true}$, then $\mathrm{nearcollision}(m \cdot P, t, P, 2 \cdot P)$.*

Proof. The goal follows from $\mathrm{periodicunitdefectforcesnearcollision}(T, P, m, t)$. □

Theorem 2.3. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then if $T \leq N$, then there exists a natural number t such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + P \wedge \mathrm{centralcollision}(m \cdot P, t, 2 \cdot P)$.*

Proof. The goal follows from $\mathrm{periodiccentralcollisionwindow}(T, P, m, N)$. □

Theorem 2.4. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then there exists a natural number t such that $N \leq t$ and $\mathrm{centralcollision}(m \cdot P, t, 2 \cdot P)$.*

Proof. The goal follows from $\mathrm{periodiccentralcollisions}(T, P, m, N)$. □

Theorem 2.5. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then $\mathrm{recurrentcentralcollision}(m \cdot P, 2 \cdot P)$.*

Proof. The goal follows from $\mathrm{periodicrecurrentcentralcolumn}(T, P, m)$. □

Theorem 2.6. *For every natural number Q and natural number b and natural number r, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b+r) =$ true*, then there exists a natural number s such that $2 \cdot s \leq r+1$ and $s \leq r \wedge$*leftcollision$(Q, b+s, r+1)$.

Proof. The goal follows from blackwindowforcesleftcollision(Q, b, r). □

Theorem 2.7. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then there exists a natural number t such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge$* leftcollision$(m \cdot P, t, P)$.

Proof. The goal follows from periodicleftcollisionwindow(T, P, m, N). □

Theorem 2.8. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then there exists a natural number t such that $N \leq t$ and* leftcollision$(m \cdot P, t, P)$.

Proof. The goal follows from periodicleftcollisions(T, P, m, N). □

Theorem 2.9. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then* recurrentleftcollision$(m \cdot P, P)$.

Proof. The goal follows from periodicrecurrentleftcolumn(T, P, m). □

Theorem 2.10. *For every natural number Q and natural number t and natural number k,* collisioncode$(Q, t, k) \neq 0$ *if and only if* collisionbit$(Q, t, k) =$ true.

Proof. The goal follows from collisioncodenonzero(Q, t, k). □

Theorem 2.11. *For every natural number Q and natural number t and natural number k,* collisioncode$(Q, t, k) \leq 6$.

Proof. The goal follows from collisioncodebound(Q, t, k). □

Theorem 2.12. *For every natural number Q and natural number t and natural number k, if $0 < k$, then* collisioncode$(Q, t, k) = 1 \leftrightarrow$ leftcolour$(Q, t, k) =$ $1 \wedge$leftcolour$(Q, t, k-1) = 2$ *and* (collisioncode$(Q, t, k) = 2 \leftrightarrow$ leftcolour$(Q, t, k) =$ $2 \wedge$leftcolour$(Q, t, k-1) = 1) \wedge$(collisioncode$(Q, t, k) = 3 \leftrightarrow$ leftcolour$(Q, t, k) =$ $1 \wedge$leftcolour$(Q, t, k-1) = 3) \wedge$(collisioncode$(Q, t, k) = 4 \leftrightarrow$ leftcolour$(Q, t, k) =$ $3 \wedge$leftcolour$(Q, t, k-1) = 1) \wedge$(collisioncode$(Q, t, k) = 5 \leftrightarrow$ leftcolour$(Q, t, k) =$ $2 \wedge$leftcolour$(Q, t, k-1) = 3) \wedge$(collisioncode$(Q, t, k) = 6 \leftrightarrow$ leftcolour$(Q, t, k) =$ $3 \wedge$ leftcolour$(Q, t, k - 1) = 2)$.

Proof. The goal follows from collisioncodecolours(Q, t, k). □

Theorem 2.13. *For every natural number Q and natural number t and natural number k,* advancingatdepth(Q, t, k) *if and only if* collisioncode$(Q, t, k) = 3$ *or* collisioncode$(Q, t, k) = 5$.

Proof. The goal follows from advancingcollisioncodes(Q, t, k). □

Theorem 2.14. *For every natural number Q and natural number t and natural number k, if* collisioncode$(Q, t, k) \neq 0$*, then* leftdefect$(Q, t + 1, k) =$ leftdefect$(Q, t, k + 1)$.

Proof. The goal follows from collisiontransmitsleft(Q, t, k). □

Theorem 2.15. *For every natural number Q and natural number t and natural number k, if* collisioncode$(Q, t, k) \neq 0$*, then* leftgatedifference$(Q, t, k) =$ false.

Proof. The goal follows from collisiongatesagree(Q, t, k). □

Theorem 2.16. *For every natural number Q and natural number t and natural number k, if $2 \leq k$, then* collisioncode$(Q, t, k) = 1 \to$ leftdefect$(Q, t+1, k-1) =$ leftcell$(t, k-2)$ *and* (collisioncode$(Q, t, k) = 2 \to$ leftdefect$(Q, t+1, k-1) =$ leftcell$(t+Q, k-2)) \wedge$(collisioncode$(Q, t, k) = 3 \to$ leftdefect$(Q, t+1, k-1) =$ true$) \wedge$ (collisioncode$(Q, t, k) = 4 \to$ leftdefect$(Q, t + 1, k - 1) =$ not(leftcell$(t + Q, k - 2))) \wedge$ (collisioncode$(Q, t, k) = 5 \to$ leftdefect$(Q, t + 1, k - 1) =$ true$) \wedge$ (collisioncode$(Q, t, k) = 6 \to$ leftdefect$(Q, t + 1, k - 1) =$ not(leftcell$(t, k - 2))$).

Proof. The goal follows from collisionrighttransporttable(Q, t, k). □

Theorem 2.17. *For every natural number Q and natural number t and natural number k, if $0 < k$, then* collisioncode$(Q, t, k) = 3 \to$ leftcolour$(Q, t + 1, k-1) = 2$ *and if* collisioncode$(Q, t, k) = 5$*, then* leftcolour$(Q, t+1, k-1) =$ 1.

Proof. The goal follows from advancingrightcolourtable(Q, t, k). □

Theorem 2.18. *For every natural number Q and natural number t and natural number k,* leftdefect$(Q, t + 1, k) =$ not(leftdefect$(Q, t, k + 1)$) *if and only if* leftgatedifference$(Q, t, k) =$ true.

Proof. The goal follows from gatedifferenceiffflip(Q, t, k). □

Theorem 2.19. *For every natural number Q and natural number t and natural number k, if* leftdefect$(Q, t, k+1) =$ true*, then* leftdefect$(Q, t+1, k) =$ false *if and only if* leftgatedifference$(Q, t, k) =$ true*.*

Proof. The goal follows from incomingdefecterasediff(Q, t, k). □

Theorem 2.20. *For every natural number Q and natural number t and natural number k, if* leftdefect$(Q, t, k+1) =$ true*, then if* leftdefect$(Q, t+1, k) =$ false*, then* collisioncode$(Q, t, k) = 0$*.*

Proof. The goal follows from erasurehasnocollision(Q, t, k). □

Theorem 2.21. *For every natural number Q and natural number k, if $0 < k$, then* leftcolour$(Q, 0, k) = 0$ *or* leftcolour$(Q, 0, k) = 2$*.*

Proof. The goal follows from seedleftcolours(Q, k). □

Theorem 2.22. *For every natural number Q and natural number k, if $2 \leq k$, then* collisioncode$(Q, 0, k) = 0$*.*

Proof. The goal follows from seedleftcollisionzero(Q, k). □

Theorem 2.23. collisioncode$(1, 1, 2) = 5$*.*

Proof. The goal follows from seedadvancingcollisionbirth. □

Theorem 2.24. leftdefect$(3, 0, 3) =$ true *and* leftdefect$(3, 1, 2) =$ false $\wedge$ leftgatedifference$(3, 0, 2) =$ true*.*

Proof. The goal follows from seedgateerasure. □

Theorem 2.25. *For every natural number Q and natural number t and natural number k, if* leftdefect$(Q, t, k) =$ true*, then if* center$(t) =$ true*, then if* center$(t + Q) =$ true*, then* advancingleftcone(Q, t, k)*.*

Proof. The goal follows from leftdefectblackcone(Q, t, k). □

Theorem 2.26. *For every natural number Q and natural number b and natural number r, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b+r) =$ true*, then there exists a natural number s such that $2 \cdot s \leq r+1$ and $s \leq r \wedge$*advancingleftcollision$(Q, b+s, r+1)$*.*

Proof. The goal follows from blackwindowforcesadvancingcollision(Q, b, r). □

Theorem 2.27. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then there exists a natural number t such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge$* advancingleftcollision$(m \cdot P, t, P)$.

Proof. The goal follows from periodicadvancingleftcollisionwindow(T, P, m, N). □

Theorem 2.28. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then* recurrentadvancingleftcollision$(m \cdot P, P)$.

Proof. The goal follows from periodicrecurrentadvancingleftcolumn(T, P, m). □

Theorem 2.29. *For every natural number Q and natural number t and natural number k,* leftdefect$(Q, t, k) =$ false *if and only if* leftcolour$(Q, t, k) = 0$ *or* leftcolour$(Q, t, k) = 3$.

Proof. The goal follows from leftdefectfalsecolours(Q, t, k). □

Theorem 2.30. *For every natural number Q and natural number t and natural number k,* leftdefect$(Q, t, k) =$ true *if and only if* leftcolour$(Q, t, k) = 1$ *or* leftcolour$(Q, t, k) = 2$.

Proof. The goal follows from leftdefecttruecolours(Q, t, k). □

Theorem 2.31. *For every natural number Q and natural number t and natural number k, if $t < k$, then* leftcolour$(Q, t, k) = 0$ *or* leftcolour$(Q, t, k) = 2$.

Proof. The goal follows from leftcoloursoutsideseedcone(Q, t, k). □

Theorem 2.32. *For every natural number Q and natural number t and natural number k, if* collisioncode$(Q, t, k) \neq 0$*, then $k \leq t + 1$.*

Proof. The goal follows from collisioncodesupport(Q, t, k). □

Theorem 2.33. *For every natural number Q and natural number t and natural number k,* gatecode$(Q, t, k) \neq 0$ *if and only if* leftgatedifference$(Q, t, k) =$ true.

Proof. The goal follows from gatecodenonzero(Q, t, k). □

Theorem 2.34. *For every natural number Q and natural number t and natural number k,* $\mathrm{gatecode}(Q, t, k) \leq 6$.

Proof. The goal follows from $\mathrm{gatecodebound}(Q, t, k)$. □

Theorem 2.35. *For every natural number Q and natural number t and natural number k, if $0 < k$, then* $\mathrm{gatecode}(Q, t, k) = 1 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 1 \wedge \mathrm{leftcolour}(Q, t, k-1) = 0$ *and* $(\mathrm{gatecode}(Q, t, k) = 2 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 0 \wedge \mathrm{leftcolour}(Q, t, k-1) = 1) \wedge (\mathrm{gatecode}(Q, t, k) = 3 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 1 \wedge \mathrm{leftcolour}(Q, t, k-1) = 1) \wedge (\mathrm{gatecode}(Q, t, k) = 4 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 2 \wedge \mathrm{leftcolour}(Q, t, k-1) = 0) \wedge (\mathrm{gatecode}(Q, t, k) = 5 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 0 \wedge \mathrm{leftcolour}(Q, t, k-1) = 2) \wedge (\mathrm{gatecode}(Q, t, k) = 6 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 2 \wedge \mathrm{leftcolour}(Q, t, k-1) = 2)$.

Proof. The goal follows from $\mathrm{gatecodecolours}(Q, t, k)$. □

Theorem 2.36. *For every natural number Q and natural number t and natural number k, if* $\mathrm{leftdefect}(Q, t, k+1) = \mathrm{true}$*, then if* $\mathrm{leftdefect}(Q, t+1, k) = \mathrm{false}$*, then* $\mathrm{erasureevent}(Q, t, k)$.

Proof. The goal follows from $\mathrm{erasureeventofstep}(Q, t, k)$. □

Theorem 2.37. *For every natural number Q and natural number t and natural number k, if* $\mathrm{erasureevent}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+1, k) = 3$ *if and only if* $\mathrm{gatecode}(Q, t, k) \leq 3$ *and* $\mathrm{leftcolour}(Q, t, k+1) = 2$ *or* $4 \leq \mathrm{gatecode}(Q, t, k)$ *and* $\mathrm{leftcolour}(Q, t, k+1) = 1$.

Proof. The goal follows from $\mathrm{erasureblackphaseiff}(Q, t, k)$. □

Theorem 2.38. *For every natural number Q and natural number t and natural number k, if $t+1 < k$, then if* $\mathrm{erasureevent}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+1, k) = 0$.

Proof. The goal follows from $\mathrm{erasureoutsideseedconewhite}(Q, t, k)$. □

Theorem 2.39. *For every natural number Q and natural number t and natural number k, if $2 \leq k$, then if* $\mathrm{advancingatdepth}(Q, t, k)$*, then if* $\mathrm{leftdefect}(Q, t+k, 0) = \mathrm{false}$*, then there exists a natural number j such that* $\mathrm{firstpatherasure}(Q, t, k, j)$.

Proof. The goal follows from $\mathrm{advancingsourcefirsterasure}(Q, t, k)$. □

Theorem 2.40. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq N$, then there exists a natural number u such that there exists a natural number d such that $N \leq u$ and $u + 1 \leq 2 \cdot N + m \cdot P + 3 \cdot P \wedge d + 2 \leq P \wedge$* $\mathrm{erasureevent}(m \cdot P, u, d)$.

Proof. The goal follows from periodicerasurewindow(T, P, m, N). □

Theorem 2.41. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P), *then* recurrentsourceerasure$(m \cdot P, P)$.

Proof. The goal follows from periodicrecurrentsourceerasure(T, P, m). □

Theorem 2.42. noperiod *if and only if* norecurrentcentralcollision.

Proof. The goal follows from noperiodiffnorecurrentcentralcollision. □

Theorem 2.43. norowcenterperiod *if and only if* norecurrentcentralcollision.

Proof. The goal follows from norowcenterperiodiffnorecurrentcentralcollision. □

Theorem 2.44. problemone *if and only if* norecurrentcentralcollision.

Proof. The goal follows from problemoneiffnorecurrentcentralcollision. □

Theorem 2.45. *If* norecurrentcentralcollision, *then* noperiod.

Proof. Assume h. The goal follows from noperiodofnorecurrentcentralcollision(h). □

Theorem 2.46. *For every natural number T and natural number P, if $0 < P$, then if* norecurrentcentralcollision, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofnorecurrentcentralcollision(T, P). □

Theorem 2.47. *If* norecurrentcentralcollision, *then* problemone.

Proof. Assume h. The goal follows from noteventuallyperiodicofnorecurrentcentralcollision(h). □

Theorem 2.48. *For every natural number b and natural number d and natural number i,* pairphase(wittphi(b), wittxor(wittphi(d), wittshl(crosscarry$(b, d), 1)), i) =$ bitvalue(rowbit(wittphi(wittxor$(b, d)), i)) -$ bitvalue(rowbit(wittphi$(b), i)$).

Proof. The goal follows from pairphasestep(b, d, i). □

Theorem 2.49. *For every natural number Q and natural number t and natural number k,* alignedphase$(Q, t, t + k + Q) =$ leftphase(Q, t, k).

Proof. The goal follows from alignedphaseleft(Q, t, k). □

Theorem 2.50. *For every natural number Q and natural number t and natural number k,* leftphase$(Q, t, k) = 0$ *if and only if* leftdefect$(Q, t, k) =$ false*.*

Proof. The goal follows from phasezeroiff(Q, t, k). □

Theorem 2.51. *For every natural number Q and natural number t and natural number k,* leftphase$(Q, t, k) = -1$ *or* leftphase$(Q, t, k) = 1$ *if and only if* leftdefect$(Q, t, k) =$ true*.*

Proof. The goal follows from phaseunitiff(Q, t, k). □

Theorem 2.52. *For every natural number Q and natural number t and natural number k,* leftphase$(Q, t+1, k) =$ leftphase$(Q, t, k+1)+$leftphaseflux(Q, t, k).

Proof. The goal follows from leftphasebalance(Q, t, k). □

Theorem 2.53. *For every natural number Q and natural number t and natural number k, if* collisioncode$(Q, t, k) \neq 0$*, then* leftphaseflux$(Q, t, k) = -2 \cdot$ leftphase$(Q, t, k + 1)$*.*

Proof. The goal follows from collisionphaseflip(Q, t, k). □

Theorem 2.54. *For every natural number Q and natural number t and natural number k, if $0 < k$, then if* advancingatdepth(Q, t, k)*, then* leftphase$(Q, t+1, k-1) = -$ leftphase(Q, t, k)*.*

Proof. The goal follows from advancingphaseflip(Q, t, k). □

Theorem 2.55. *For every natural number Q and natural number t and natural number k,* gatecode$(Q, t, k) = 1 \to$ leftgatephase$(Q, t, k) = -1$ *and* (gatecode$(Q, t, k) = 2 \to$ leftgatephase$(Q, t, k) = -1) \wedge$ (gatecode$(Q, t, k) = 3 \to$ leftgatephase$(Q, t, k) = -1) \wedge$(gatecode$(Q, t, k) = 4 \to$ leftgatephase$(Q, t, k) = 1) \wedge$(gatecode$(Q, t, k) = 5 \to$ leftgatephase$(Q, t, k) = 1) \wedge$(gatecode$(Q, t, k) = 6 \to$ leftgatephase$(Q, t, k) = 1)$.

Proof. The goal follows from gatephasecases(Q, t, k). □

Theorem 2.56. *For every natural number Q and natural number t and natural number k, if* erasureevent(Q, t, k)*, then* leftphase$(Q, t + 1, k) = 0$ *and* leftphaseflux$(Q, t, k) = -$ leftphase$(Q, t, k+1) \wedge$ (leftphase$(Q, t, k+1) = -1 \vee$ leftphase$(Q, t, k + 1) = 1)$*.*

Proof. The goal follows from erasurephasebalance(Q, t, k). □

Theorem 2.57. *For every natural number Q and natural number t and natural number k, if* erasureevent(Q, t, k)*, then* leftcolour$(Q, t+1, k) = 3 \leftrightarrow$ leftgatephase$(Q, t, k) \cdot$leftphase$(Q, t, k+1) = -1$ *and* leftcolour$(Q, t+1, k) = 0 \leftrightarrow$ leftgatephase$(Q, t, k) \cdot$ leftphase$(Q, t, k+1) = 1$.

Proof. The goal follows from erasureresetproduct(Q, t, k). □

Theorem 2.58. *For every natural number Q and natural number t and natural number k and natural number n,* pathphase$(Q, t, k, n) =$ leftphase$(Q, t, k)+$ pathflux(Q, t, k, n).

Proof. The goal follows from pathphasebalance(Q, t, k, n). □

Theorem 2.59. *For every natural number Q and natural number t, if $0 < Q$, then* leftphase$(Q, t, t + Q) = 1$.

Proof. The goal follows from seedleftphase(Q, t). □

Theorem 2.60. *For every natural number Q and natural number t, if $0 < Q$, then* leftphase$(Q, t + (t + Q), 0) = 1 +$ pathflux$(Q, t, t + Q, t + Q)$.

Proof. The goal follows from seedcenterphasebalance(Q, t). □

Theorem 2.61. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then* pathflux$(m \cdot P, N, N + m \cdot P, N + m \cdot P) = -1$.

Proof. The goal follows from periodicseedflux(T, P, m, N). □

Theorem 2.62. leftphase$(3, 0, 3) = 1$ *and* leftphase$(3, 1, 2) = 0 \wedge$leftphaseflux$(3, 0, 2) = -1 \wedge$ leftgatephase$(3, 0, 2) = 1$.

Proof. The goal follows from seedphasecancellation. □

Theorem 2.63. *For every natural number Q and natural number t and natural number k and natural number s, if* rightmostleft(Q, t, k)*, then if $s < k$, then if* centercleanthrough(Q, t, s)*, then* rightmostleft$(Q, t + s, k - s)$.

Proof. The goal follows from rightmostleftprofile(Q, t, k, s). □

Theorem 2.64. *For every natural number Q and natural number t and natural number k, if* leadingadvance(Q, t, k)*, then if* centercleanthrough(Q, t, k)*, then* firstpatherasure$(Q, t, k, k - 2)$ *and* erasureevent$(Q, t + (k - 1), 0) \wedge (\text{gatecode}(Q, t + (k-1), 0) = 2 \vee \text{gatecode}(Q, t + (k-1), 0) = 5) \wedge \text{center}(t + (k - 1)) = \text{false}$.

Proof. The goal follows from leadingfirsterasure(Q, t, k). □

Theorem 2.65. *For every natural number Q and natural number b and natural number r, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b + r) =$ true*, then there exists a natural number s such that there exists a natural number k such that $2 \cdot s \leq r + 1$ and $s \leq r \wedge k + s \leq r + 1 \wedge$* leadingadvance$(Q, b + s, k)$.

Proof. The goal follows from blackwindowforcesleadingsource(Q, b, r). □

Theorem 2.66. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then there exists a natural number t such that there exists a natural number k such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge k \leq P \wedge$* leadingadvance$(m \cdot P, t, k)$.

Proof. The goal follows from periodicleadingsourcewindow(T, P, m, N). □

Theorem 2.67. *For every natural number T and natural number P and natural number m and natural number N, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $T \leq N$, then there exists a natural number t such that there exists a natural number k such that $N \leq t$ and $t+1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge k \leq P \wedge$* leadingadvance$(m \cdot P, t, k) \wedge$ firstpatherasure$(m \cdot P, t, k, k - 2) \wedge (\text{gatecode}(m \cdot P, t + (k-1), 0) = 2 \vee \text{gatecode}(m \cdot P, t + (k-1), 0) = 5)$.

Proof. The goal follows from periodicleadingerasurewindow(T, P, m, N). □

Theorem 2.68. *For every natural number T and natural number P and natural number m, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then* recurrentleadingerasure$(m \cdot P, P)$.

Proof. The goal follows from periodicrecurrentleadingerasure(T, P, m). □

Theorem 2.69. *For every natural number Q and natural number R and natural number t and natural number k,* leftphase$(Q+R, t, k) =$ leftphase$(Q, t, k)+$ leftphase$(R, t + Q, k)$.

Proof. The goal follows from leftphaselagadd(Q, R, t, k). □

Theorem 2.70. *For every natural number Q and natural number R and natural number t and natural number k and natural number n,* $\mathrm{pathflux}(Q + R, t, k, n) = \mathrm{pathflux}(Q, t, k, n) + \mathrm{pathflux}(R, t + Q, k, n)$.

Proof. The goal follows from $\mathrm{pathfluxlagadd}(Q, R, t, k, n)$. □

Theorem 2.71. *For every natural number Q and natural number t and natural number k, if* $\mathrm{leadingadvance}(Q, t, k)$*, then if* $\mathrm{centercleanthrough}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+k, 0) = 3 \leftrightarrow \mathrm{leftgatephase}(Q, t+(k-1), 0) \cdot (\mathrm{leftphase}(Q, t, k) + \mathrm{pathflux}(Q, t, k, k-1)) = -1$ *and* $\mathrm{leftcolour}(Q, t+k, 0) = 0 \leftrightarrow \mathrm{leftgatephase}(Q, t + (k-1), 0) \cdot (\mathrm{leftphase}(Q, t, k) + \mathrm{pathflux}(Q, t, k, k-1)) = 1$.

Proof. The goal follows from $\mathrm{leadingphasereset}(Q, t, k)$. □

Theorem 2.72. *For every natural number Q and natural number t, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{leftdefect}(Q, t + 1, 0) = \mathrm{false}$*, then* $\mathrm{erasureevent}(Q, t, 0)$ *if and only if* $\mathrm{gatecode}(Q, t, 0) = 2$ *or* $\mathrm{gatecode}(Q, t, 0) = 5$.

Proof. The goal follows from $\mathrm{centererasureiffgate}(Q, t)$. □

Theorem 2.73. *For every natural number Q and natural number b and natural number n and natural number i and natural number j and natural number k, if* $\mathrm{commonwhiteblock}(Q, b, n)$*, then if $i \leq j$, then if $j \leq k$, then if $k \leq n$, then if* $\mathrm{gatecode}(Q, b + i, 0) = 2$ *or* $\mathrm{gatecode}(Q, b + i, 0) = 5$*, then if* $\mathrm{gatecode}(Q, b + k, 0) = 2$ *or* $\mathrm{gatecode}(Q, b + k, 0) = 5$*, then* $\mathrm{gatecode}(Q, b + j, 0) = \mathrm{gatecode}(Q, b + i, 0)$.

Proof. The goal follows from $\mathrm{whiteblockgateconvex}(Q, b, n, i, j, k)$. □

Theorem 2.74. *For every natural number Q and natural number b and natural number n and natural number i and natural number j and natural number k, if* $\mathrm{commonwhiteblock}(Q, b, n)$*, then if* $\mathrm{centercleanthrough}(Q, b, n+1)$*, then if $i \leq j$, then if $j \leq k$, then if $k \leq n$, then if* $\mathrm{erasureevent}(Q, b + i, 0)$*, then if* $\mathrm{erasureevent}(Q, b + k, 0)$*, then* $\mathrm{erasureevent}(Q, b + j, 0)$ *and* $\mathrm{gatecode}(Q, b + j, 0) = \mathrm{gatecode}(Q, b + i, 0)$.

Proof. The goal follows from $\mathrm{whiteblockerasureconvex}(Q, b, n, i, j, k)$. □

Theorem 2.75. $\mathrm{leadingadvance}(4, 5, 2)$ *and* $\mathrm{centercleanthrough}(4, 5, 2) \wedge \mathrm{gatecode}(4, 6, 0) = 5$.

Proof. The goal follows from seedleadinggatefive. □

Theorem 2.76. $\mathrm{leadingadvance}(7, 7, 4)$ *and* $\mathrm{centercleanthrough}(7, 7, 4) \wedge \mathrm{gatecode}(7, 10, 0) = 2$.

Proof. The goal follows from seedleadinggatetwo. □

Theorem 2.77. leadingadvance$(11, 4, 4)$ *and* centercleanthrough$(11, 4, 4) \wedge$ gatecode$(11, 7, 0) = 5 \wedge$ leftcolour$(11, 8, 0) = 3$.

Proof. The goal follows from seedleadingblackreset. □

Theorem 2.78. *For every truth value incomingBit and truth value laterBit and integer incoming and integer gate,* recoverphase$(incomingBit, laterBit,$ transferphase$(incomingBi$ *incoming.*

Proof. The goal follows from transferrecover$(incomingBit, laterBit, incoming, gate)$. □

Theorem 2.79. *For every truth value incomingBit and truth value laterBit and integer incoming and integer gate,* transferphase$(incomingBit, laterBit, incoming, gate) =$ 0 *if and only if incoming* $=$ recoverphase$(incomingBit, laterBit, 0, gate)$.

Proof. The goal follows from transferkernel$(incomingBit, laterBit, incoming, gate)$. □

Theorem 2.80. *For every natural number Q and natural number t and natural number k,* leftphase$(Q, t{+}1, k) =$ transferphase(leftcell$(t, k{+}1)$, latergate(Q, t, k), leftphase$(Q, t, k{+}1)$, leftgatephase(Q, t, k)).

Proof. The goal follows from phasetransfer(Q, t, k). □

Theorem 2.81. *For every natural number Q and natural number t and natural number k,* leftphase$(Q, t, k{+}1) =$ recoverphase(leftcell$(t, k{+}1)$, latergate(Q, t, k), leftphase$(Q, t{+}1, k)$, leftgatephase(Q, t, k)).

Proof. The goal follows from phaserecover(Q, t, k). □

Theorem 2.82. *For every natural number Q and natural number t and natural number k, if* erasureevent(Q, t, k)*, then not* leftgatephase$(Q, t, k) = 0$ *and* leftphase$(Q, t, k{+}1) =$ recoverphase(leftcell$(t, k{+}1)$, latergate$(Q, t, k), 0,$ leftgatephase(Q, t, k)).

Proof. The goal follows from erasurerecovery(Q, t, k). □

Theorem 2.83. *For every natural number T and natural number P and natural number m and natural number N and natural number g, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $g = 2$ or $g = 5$, then there exists a natural number t such that $N \leq t$ and* erasureevent$(m \cdot P, t, 0) \wedge$ gatecode$(m \cdot P, t, 0) = g$.

Proof. The goal follows from periodicerasureorientation(T, P, m, N, g). □

Theorem 2.84. *For every natural number Q and natural number t and natural number k, if* leadingadvance(Q, t, k)*, then if* centercleanthrough(Q, t, k)*, then there exists a natural number s such that $s + 1 < k$ and* center$(t + s) =$ true*.*

Proof. The goal follows from leadingblackbeforearrival(Q, t, k). □

Theorem 2.85. *For every natural number Q and natural number b and natural number r, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* whitecentergap(b, r)*, then if* leftadjacentdefect$(Q, b+ r) =$ true*, then there exists a natural number k such that $k \leq r + 1$ and* leadingadvance(Q, b, k)*.*

Proof. The goal follows from blackgapforcesinitialleading(Q, b, r). □

Theorem 2.86. *For every natural number Q and natural number t, if* leadingadvance$(Q, t, 2)$*, then if* centercleanthrough$(Q, t, 2)$*, then* center$(t) =$ true*.*

Proof. The goal follows from leadingdepthtwoblack(Q, t). □

Theorem 2.87. *For every natural number Q and natural number b and natural number r and natural number g, if $0 < r$, then if* centerwindowagree$(b, b+ Q, r + 1)$*, then if* center$(b) =$ true*, then if* whitecentergap(b, r)*, then if* erasureevent$(Q, b + r, 0)$*, then if* gatecode$(Q, b + r, 0) = g$*, then there exists a natural number k such that $k \leq r + 1$ and* leadingadvance$(Q, b, k) \wedge$ firstpatherasure$(Q, b, k, k - 2) \wedge$ gatecode$(Q, b + (k - 1), 0) = g$*.*

Proof. The goal follows from blackgaperasureorigin(Q, b, r, g). □

Theorem 2.88. *For every natural number T and natural number P and natural number m and natural number N and natural number g, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $g = 2$ or $g = 5$, then there exists a natural number b such that there exists a natural number k such that $N \leq b$ and $k \leq P \wedge$* center$(b) =$ true $\wedge$ whitecentergap$(b, k - 1) \wedge$ leadingadvance$(m \cdot P, b, k) \wedge$ firstpatherasure$(m \cdot P, b, k, k-2) \wedge$ gatecode$(m \cdot P, b + (k - 1), 0) = g$*.*

Proof. The goal follows from periodicblackerasureorigin(T, P, m, N, g). □

Theorem 2.89. *For every natural number t,* rightcell$(t+1, 1) =$ bxor(center(t), bxor(rightcell$(t, 1)$, right

Proof. The goal follows from rightimportstep(t). □

Theorem 2.90. *For every natural number Q and natural number t, if* $\mathrm{center}(t+Q) = \mathrm{center}(t)$*, then* $\mathrm{rightdefect}(Q, t+1, 1) = \mathrm{bxor}(\mathrm{rightdefect}(Q, t, 1), \mathrm{rightimportdifference}(Q, t))$.

Proof. The goal follows from $\mathrm{rightdefectimportbalance}(Q, t)$. □

Theorem 2.91. *For every natural number Q and natural number t, if* $\mathrm{center}(t) = \mathrm{true}$*, then if* $\mathrm{center}(t+Q) = \mathrm{true}$*, then if* $\mathrm{rightdefect}(Q, t, 1) = \mathrm{true}$*, then if* $\mathrm{rightdefect}(Q, t+1, 1) = \mathrm{true}$*, then* $\mathrm{rightphase}(Q, t+1, 1) = -\mathrm{rightphase}(Q, t, 1)$.

Proof. The goal follows from $\mathrm{blackresetsurvival}(Q, t)$. □

Theorem 2.92. *For every natural number Q and natural number t, if* $\mathrm{center}(t) = \mathrm{true}$*, then if* $\mathrm{center}(t+Q) = \mathrm{true}$*, then if* $\mathrm{rightdefect}(Q, t, 1) = \mathrm{false}$*, then if* $\mathrm{rightdefect}(Q, t+1, 1) = \mathrm{true}$*, then* $\mathrm{rightcell}(t, 1) = \mathrm{false}$ *and* $\mathrm{rightcell}(t+Q, 1) = \mathrm{false} \wedge \mathrm{rightdefect}(Q, t, 2) = \mathrm{true} \wedge \mathrm{rightphase}(Q, t+1, 1) = -\mathrm{rightphase}(Q, t, 2)$.

Proof. The goal follows from $\mathrm{blackresetimport}(Q, t)$. □

Theorem 2.93. *For every natural number T and natural number P and natural number m and natural number N and truth value $before$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then there exists a natural number t such that $N \leq t$ and* $\mathrm{rightimport}(t) = before \wedge \mathrm{rightimport}(t + m \cdot P) = \mathrm{not}(before) \wedge \mathrm{rightdefect}(m \cdot P, t+1, 1) = \mathrm{not}(\mathrm{rightdefect}(m \cdot P, t, 1))$.

Proof. The goal follows from $\mathrm{periodicrightimportorientation}(T, P, m, N, before)$. □

Theorem 2.94. *For every natural number Q and natural number t, if* $\mathrm{center}(t+Q) = \mathrm{center}(t)$*, then if* $\mathrm{rightdefect}(Q, t, 1) = \mathrm{false}$*, then if* $\mathrm{rightdefect}(Q, t+1, 1) = \mathrm{true}$*, then* $\mathrm{rightcell}(t, 1) = \mathrm{false}$ *and* $\mathrm{rightcell}(t+Q, 1) = \mathrm{false} \wedge \mathrm{rightdefect}(Q, t, 2) = \mathrm{true} \wedge \mathrm{rightimportdifference}(Q, t) = \mathrm{true} \wedge \mathrm{rightphase}(Q, t+1, 1) = (1 - 2 \cdot \mathrm{bitvalue}(\mathrm{center}(t))) \cdot \mathrm{rightphase}(Q, t, 2)$.

Proof. The goal follows from $\mathrm{rightheadbirth}(Q, t)$. □

Theorem 2.95. *For every natural number Q and natural number $start$ and natural number $finish$, if* $\mathrm{headimportpath}(Q, start, finish)$*, then if* $\mathrm{centercleanthrough}(Q, start, finish - start)$*, then* $\mathrm{rightphase}(Q, finish, 1) = \mathrm{centersign}(start, finish - start) \cdot \mathrm{rightphase}(Q, start, 2)$.

Proof. The goal follows from $\mathrm{headimportpathphase}(Q, start, finish)$. □

Theorem 2.96. *For every natural number Q and natural number t, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{leftdefect}(Q, t+1, 0) = \mathrm{false}$*, then if* $\mathrm{center}(t) = \mathrm{false}$*, then* $\mathrm{leftphase}(Q, t, 1) = (1 - 2 \cdot \mathrm{bitvalue}(\mathrm{center}(t+1))) \cdot \mathrm{rightphase}(Q, t, 1)$.

Proof. The goal follows from whitecenterphasematching(Q, t). □

Theorem 2.97. *For every natural number T and natural number P and natural number m and natural number N and natural number g, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $g = 2$ or $g = 5$, then there exists a natural number start such that there exists a natural number b such that there exists a natural number k such that $N \leq start$ and $N \leq b \wedge k \leq P \wedge$* center$(b) =$ true $\wedge$ whitecentergap$(b, k-1) \wedge$ leadingadvance$(m \cdot P, b, k) \wedge$ firstpatherasure$(m \cdot P, b, k, k-2) \wedge$ gatecode$(m \cdot P, b + (k-1), 0) = g \wedge$headimportpath$(m \cdot P, start, b + (k-1)) \wedge$leftphase$(m \cdot P, b, k) +$ pathflux$(m \cdot P, b, k, k-1) = (1 - 2 \cdot$ bitvalue(center$(b + k))) \cdot$ centersign$(start, b + (k - 1) - start) \cdot$ rightphase$(m \cdot P, start, 2)$.

Proof. The goal follows from periodiccoupledcancellation(T, P, m, N, g). □

Theorem 2.98. *For every natural number Q and natural number b and natural number k, if* leadingadvance(Q, b, k)*, then if* centercleanthrough(Q, b, k)*, then* leftphase$(Q, b, k) +$ pathflux$(Q, b, k, k-1) = (1 - 2 \cdot$ bitvalue(center$(b + k))) \cdot$ leftgatephase$(Q, b + (k-1), 0)$.

Proof. The goal follows from leadingarrivalphaseeqgatephase(Q, b, k). □

Theorem 2.99. *For every natural number Q and natural number start and natural number $finish$, if* headimportpath$(Q, start, finish)$*, then if* centercleanthrough$(Q, start, finish - start)$*, then if* center$(finish) =$ false*, then* leftgatephase$(Q, finish, 0) =$ centersign$(start, finish - start) \cdot$rightphase$(Q, start, 2)$.

Proof. The goal follows from headimportpathgatephase$(Q, start, finish)$. □

Theorem 2.100. *For every natural number Q and natural number start and natural number $finish$, if* headimportpath$(Q, start, finish)$*, then if* centercleanthrough$(Q, start, finish - start)$*, then if* center$(finish) =$ false*, then not* leftgatephase$(Q, finish, 0) = -($centersign$(start, finish - start) \cdot$ rightphase$(Q, start, 2))$.

Proof. The goal follows from headimportpathrejectsnegativegate$(Q, start, finish)$. □

Theorem 2.101. *For every natural number T and natural number P and natural number m and natural number N and natural number g, if $0 < P$, then if $0 < m$, then if* centerperiodic(T, P)*, then if $g = 2$ or $g = 5$, then there exists a natural number start such that there exists a natural number*

b *such that there exists a natural number* k *such that* $N \leq start$ *and* $N \leq b \wedge k \leq P \wedge \mathrm{center}(b) = \mathrm{true} \wedge \mathrm{whitecentergap}(b, k-1) \wedge \mathrm{leadingadvance}(m \cdot P, b, k) \wedge \mathrm{firstpatherasure}(m \cdot P, b, k, k-2) \wedge \mathrm{gatecode}(m \cdot P, b+(k-1), 0) = g \wedge \mathrm{headimportpath}(m \cdot P, start, b+(k-1)) \wedge \mathrm{leftgatephase}(m \cdot P, b+(k-1), 0) = \mathrm{centersign}(start, b+(k-1)-start) \cdot \mathrm{rightphase}(m \cdot P, start, 2)$.

Proof. The goal follows from $\mathrm{periodiccoupledgatephasesign}(T, P, m, N, g)$. □

Theorem 2.102. *For every natural number* T *and natural number* P *and natural number* m *and natural number* N*, if* $0 < P$*, then if* $0 < m$*, then if* $\mathrm{centerperiodic}(T, P)$*, then if* $T \leq N$*, then* $\mathrm{pathflux}(m \cdot P + m \cdot P, N, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = -1$ *and* $\mathrm{pathflux}(m \cdot P, N, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = 0 \wedge \mathrm{pathflux}(m \cdot P, N + m \cdot P, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = -1$.

Proof. The goal follows from $\mathrm{periodicdoubledseedfluxsplit}(T, P, m, N)$. □

Theorem 2.103. $\mathrm{leadingadvance}(11, 4, 4)$ *and* $\mathrm{centercleanthrough}(11, 4, 4) \wedge \mathrm{headimportpath}(11, 6, 7) \wedge \mathrm{centercleanthrough}(11, 6, 1) \wedge \mathrm{leftgatephase}(11, 7, 0) = 1 \wedge \mathrm{centersign}(6, 1) = 1 \wedge \mathrm{rightphase}(11, 6, 2) = 1$.

Proof. The goal follows from seedcoupledgatesign. □